

A Field Study of the Sunkoshi Hydropower Station: Generation, Transmission, and Operational Challenges

MANISH KARN¹

¹Department of Electrical and Electronics Engineering, School of Engineering, Kathmandu University, Dhulikhel, Kavre, Nepal (e-mail: manishkarn.mk@gmail.com)

Corresponding author: Manish Karn (e-mail: manishkarn.mk@gmail.com).

This report is based on an academic field visit undertaken as part of the undergraduate curriculum of the Department of Electrical and Electronics Engineering, Kathmandu University.

ABSTRACT This paper presents a field study of the Sunkoshi Hydropower Station, a run-of-river plant located in Lamoshangu, Sindhupalchowk district, Nepal, conducted as part of an undergraduate field visit by the Department of Electrical and Electronics Engineering, Kathmandu University. The station, commissioned in January 1972 through the cooperation of the Water Conservancy and Electric Power Ministry of the People's Republic of China and the Government of Nepal, has an installed capacity of 10.05 MW across three Francis turbine-generator units. The study documents the complete generation chain of the plant, including the intake and head works, the open canal and penstock conveyance system, the powerhouse and its electromechanical equipment, the electromechanical governing and control system, and the switchyard, protection, and transmission arrangement that delivers power at 66 kV toward Kathmandu and at 11 kV to surrounding districts. Particular attention is given to the operational challenges faced by the plant, including seasonal discharge variation, sedimentation and solid-waste intrusion at the intake, and the structural damage caused by landslides and the 2015 earthquake, which together reduced the station's output to approximately 60% of its rated capacity for an extended period. The study concludes by relating the observed engineering practices to the broader context of hydropower development, local electrification, and irrigation in Nepal, and highlights the value of field-based learning in bridging classroom theory with practical power system operation.

INDEX TERMS Electrical power generation, Francis turbine, hydroelectric power plants, power distribution, power transmission, run-of-river power plant, Sunkoshi Hydropower Station.

I. INTRODUCTION

HYDROPOWER remains the dominant source of electrical energy in Nepal owing to the country's steep river gradients and high annual discharge. Understanding the complete chain of hydroelectric generation, from intake to transmission, is an essential part of the training of electrical and electronics engineering students. As part of this training, students from the Department of Electrical and Electronics Engineering, Kathmandu University, undertook a field visit to the Sunkoshi Hydropower Station, located approximately 81 km east of Kathmandu in Lamoshangu, Sindhupalchowk district. The objective of the visit was to observe, first-hand, the generation, transmission, and distribution of hydroelectricity, and to relate the theoretical concepts taught in the

classroom to an operating power plant.

This paper reports the observations made during the visit. Section II summarizes the salient features of the plant. Section III outlines the scope of the site visit. Section IV presents the detailed discussion and observations covering the headworks, waterways, powerhouse, governing system, and switchyard. Section V discusses the effects of landslides and the 2015 earthquake on plant operation. Section VI concludes the paper.

II. PLANT OVERVIEW

The Sunkoshi Hydropower Station is a run-of-river, daily-pondage plant with an installed capacity of 10.05 MW and an annual design generation of 70 GWh. The station com-

TABLE 1. Salient Features of the Sunkoshi Hydropower Station

Parameter	Value
Plant type	Run-of-river
Location	Lamoshangu, Sindhupalchowk
Installed capacity	10.05 MW
Design discharge	39.9 m ³ /s
Maximum net head	30.5 m
Length of canal	2.653 km
Penstock diameter	2.54 m (3 nos.)
Turbine-generator sets	3
Shaft configuration	Vertical
Turbine type	Francis (Model: HL123a-LJ-140)
Turbine output	3530 kW
Turbine speed	300 rpm
Generator type	Synchronous, 3-phase (Model: TS 325/36-20)
Generator capacity	4000 kVA
Rated voltage	6.3 kV
Rated current	361 A
Rated power factor	0.85
Transmission line	66 kV, single circuit
Project inception	End of 1968
Commissioning date	January 1972
Financed by	People's Republic of China and Government of Nepal
Project cost	NRs. 109.37 million (including transmission line)

Source: Nepal Electricity Authority [1].

prises three generating units rated at 3.35 MW each. The powerhouse was commissioned in January 1972 under the friendly cooperation of the Water Conservancy and Electric Power Ministry of the People's Republic of China and the Government of Nepal. The total project cost, including the transmission line up to Kathmandu, was approximately NRs. 109.4 million. The salient features of the plant are summarized in Table 1.

III. SITE VISIT METHODOLOGY

The field visit was carried out with the guidance of the Department of Electrical and Electronics Engineering, Kathmandu University, with the objective of studying the generation of electricity at a hydropower plant. The study of any hydropower station requires an understanding of four principal components: the head works, the canal or tunnel conveying water to the powerhouse, the powerhouse itself, and the associated mechanical and electrical equipment. Observations at the Sunkoshi station were organized around these four components, supplemented by discussions with the station's engineers regarding operational challenges and recent damage caused by natural hazards.

In December 1968, under the friendly cooperation of the Water Conservancy and Electric Power Ministry of the People's Republic of China and the Government of Nepal, construction of the Sunkoshi hydropower plant began with a planned capacity of 10.5 MW (Fig. 1). After a construction period of three years, the plant was commissioned in January 1972, with a total allocated budget of approximately NRs. 11 crore.

**FIGURE 1.** Signboard of the Sunkoshi Hydropower Station, Nepal Electricity Authority, Lamoshangu.**FIGURE 2.** Head walls of the intake structure at Lamoshangu.

IV. DISCUSSION AND OBSERVATION

A. HEAD WORKS AND WATER CONVEYANCE

The intake structure, or head works, diverts water from the Sunkoshi river into the plant's conveyance system (Fig. 2). With a gross discharge capacity of 88 m³/s from the Sunkoshi river, a diversion structure is located at Lamoshangu, comprising a 2793 m long open canal. Water is stored in a reservoir that is fed through a four-way intake from the dam (Fig. 4).

Water travels from the dam to the reservoir through an open channel (Fig. 5) with a discharge capacity of 40 m³/s and is collected in a daily regulating pond, which has a design discharge capacity of 67,000 m³/s (Fig. 6). From the regulating pond, water is passed through a four-way structure with a storage capacity of 18,000 m³ before entering three separate penstocks, each connected to one of the three turbines housed in the powerhouse (Fig. 7). Each penstock pipe has a discharge capacity of 13.3 m³/s, giving a combined capacity of 39.9 m³/s for the three pipes. Excess water in the reservoir is returned to the Sunkoshi river through a spillway (Fig. 8).



FIGURE 3. Open canal conveying water from the intake toward the reservoir.



FIGURE 4. Diversion structure (dam) at Lamoshangu.



FIGURE 5. Water conveyance channel from the dam to the reservoir.

B. POWERHOUSE AND ELECTROMECHANICAL EQUIPMENT

The powerhouse contains the mechanical and electrical equipment associated with the three turbine-generator units (Fig. 9–11). The mechanical equipment includes the turbine assemblies, governing linkages, and auxiliary pumps, while the electrical equipment consists of the generator control panels, protection relays, and metering instruments used by station operators.



FIGURE 6. Daily regulating pond (reservoir) of the Sunkoshi station.



FIGURE 7. Three penstock pipes, each connected to one of the three turbines in the powerhouse.



FIGURE 8. Spillway channel returning excess reservoir water to the Sunkoshi river.

C. TURBINE AND GENERATOR OPERATION

Each of the three turbines has a rated capacity of 3.53 MW and operates at a speed of 300 rpm on a vertically aligned shaft (Fig. 12 and Fig. 13). With a normal operating head of 30.5 m, the Sunkoshi station is classified as a medium-head plant.

To operate at full capacity, water from the reservoir is directed through each penstock pipe at its rated discharge of $13.3 \text{ m}^3/\text{s}$ (Fig. 14). As the water strikes the runner of the Francis turbine (Fig. 15), the turbine begins to rotate, driving the generator rotor, which is coupled to the same vertical shaft. The magnetic flux produced by the stationary field coil, when intersected by the rotating rotor coil, induces an electromotive force in the stator windings, generating power at 6.3 kV, 50 Hz, and a power factor of 0.85. Station operators monitor the output frequency and voltage using dedicated measurement panels (Fig. 16 and Fig. 17).



FIGURE 9. Exterior view of the Sunkoshi powerhouse.



FIGURE 10. Mechanical equipment inside the powerhouse.



FIGURE 11. Electrical control and metering equipment inside the powerhouse.



FIGURE 12. Vertical shaft alignment connecting the turbine and generator.



FIGURE 13. Turbine head assembly.

D. GOVERNING AND CONTROL SYSTEM

The plant's original manual control system was replaced approximately nine years prior to the visit with an electromechanical governor control system (Fig. 18). Under this system, variation in rotational speed caused by changes in load is corrected by regulating the discharge of water in a short response time, keeping the output voltage and frequency of the generator constant. The governing system commands a servo motor to simultaneously actuate sixteen guide vanes, which direct the flow of water through the nozzle with equal velocity onto each blade of the Francis turbine.

E. SWITCHYARD, PROTECTION, AND TRANSMISSION

For control and protection, the station uses ten circuit breakers: three for the generators, two for the power transformers, two for the station supply, and the remainder for local feeders (Fig. 19 and Fig. 20). Minimum-oil circuit breakers (MOCB), originally installed at the station, have since been replaced by vacuum circuit breakers (VCB). The 11 kV transmission line uses vacuum circuit breakers, the 66 kV transmission line uses SF₆ circuit breakers, and the local feeders use air circuit



FIGURE 14. Francis turbine units installed in the powerhouse.



FIGURE 15. Runner of the Francis turbine.



FIGURE 16. Frequency measurement instrumentation at the control panel.

breakers (ACB).

Because the reservoir does not receive adequate inflow during the dry season, the station does not run at full capacity throughout the year; the regulating pond partially compensates for this shortfall and thereby increases the operational importance of the plant (Fig. 22). The plant employs several cooling methods, including natural air cooling, forced-air cooling, forced-water cooling, and oil cooling, to minimize faults arising from excessive temperature rise.

The switchyard adjacent to the powerhouse is notable in that nearly every voltage level currently used in Nepal,



FIGURE 17. Voltage measuring devices at the control panel.



FIGURE 18. Electromechanical governor and control system, including the control cabinet, servo-driven governor motors, guide-vane linkages, and the turbine spiral casing.

with the exception of 132 kV, can be observed there. Two power transformers rated at 6.3 MVA step up the 6.3 kV generator bus to the 66 kV bus. An SF₆ circuit breaker rated at 2500 MVA connects the 66 kV bus to the transmission line supplying Bhaktapur, where 11 kV supplies received from the Khimti and Bhotekoshi hydropower stations are also stepped up to 66 kV and combined with the output of Indrawati Hydropower. A 33 kV substation located above the powerhouse supplies Dolakha through a dedicated trans-



FIGURE 19. Power transformers in the switchyard.



FIGURE 20. Circuit breakers installed in the switchyard.



FIGURE 21. Transmission lines and local feeders leaving the switchyard.



FIGURE 22. Regulating pond of the Sunkoshi Hydropower Station.

mission line, while an 11 kV substation supplies Barahabise, Chautara, and Lamoshangu, among other areas, through six feeders. A further step-down from 11 kV to 6.3 kV supplies the Sunkoshi Small Hydropower Project at Barahabise. Local supply to the powerhouse and dam area is provided at 400 V.

The switchyard is protected against lightning strikes by a dedicated lightning tower, and against touch and step potentials by specialized earthing and stone dressing. Power transformers are protected from excessive heating using oil-natural-air-natural (ONAN) and oil-natural-air-forced (ONAF) cooling arrangements. Electricity generated at the

station is distributed to the local and neighboring service areas, with the remainder transmitted at 66 kV toward Kathmandu. During the off-season, particularly at peak hours, local demand can exceed local generation, requiring power to be imported from Kathmandu; the transmission arrangement is therefore designed for bidirectional power flow.

F. LOCAL ELECTRIFICATION AND IRRIGATION

The Sunkoshi Hydropower Station has given priority to local electrification since its commissioning. The presence of the plant has directly affected part of the local irrigation network; however, water used for power generation is subsequently routed through separate canals for irrigation, which also provides employment opportunities for the local population. The station, built through the cooperation of the Water Conservancy and Electric Power Ministry of the People's Republic of China and the Government of Nepal, is currently under the protection of the Nepal Army.

V. EFFECTS OF LANDSLIDE AND EARTHQUAKE

The Sunkoshi plant requires annual maintenance following the monsoon season due to the severe damage caused by landslides. According to a joint statement issued by Nepalese agencies, two of the dam's ten iron gates were completely destroyed by a landslide, and the turbines, generators, and other equipment were also affected. Because the intake structure that previously blocked debris such as paper, plastic bottles, and other solid waste was damaged, this waste now enters the turbines directly. To prevent turbine damage, personnel are employed to continuously remove waste manually from the underground floor of the intake.

This problem could be substantially reduced if the damaged gates were repaired and if awareness campaigns were conducted among local residents, since much of the waste entering the river originates from residents using it as a disposal site. Discussions with the station's engineers during the visit confirmed that the river in this area has effectively become a local waste disposal area.

In addition to landslide damage, the Gorkha earthquake of 2015 caused further damage to the station. According to subsequent reports, landslides combined with earthquake damage blocked most of the Sunkoshi dam's gates, raising concern that the structure might be overtopped. Power generation resumed approximately 40 days after the landslide event, although the plant continued to operate at only about 60% of its normal capacity for a considerable period thereafter.

VI. CONCLUSION

This field visit to the Sunkoshi Hydropower Station provided a practical complement to classroom instruction in electrical power generation, transmission, and distribution. Direct observation of the head works, waterways, powerhouse, governing system, and switchyard clarified the engineering concepts and equipment involved in converting the flow of the Sunkoshi river into electrical energy. The visit also

exposed the operational challenges faced by an aging run-of-river plant, including seasonal discharge variation, sediment and solid-waste intrusion, and the compounding effects of landslides and earthquakes on plant availability. These observations reinforce the importance of resilient design and regular maintenance in sustaining Nepal's hydropower infrastructure, and the visit strengthened the author's interest in pursuing further work in this field.

ACKNOWLEDGMENT

The author gratefully acknowledges the support of the Department of Electrical and Electronics Engineering, Kathmandu University, in organizing the field visit on which this report is based, and in particular the guidance of Assistant Professor Basanta Raj Pokharel, Mr. Shailendra Kumar Jha, and Mr. Parash Acharya, along with the other faculty members who accompanied and supported the students during the visit and provided insight into the principles of hydroelectric power generation.

REFERENCES

- [1] Nepal Electricity Authority. Sunkoshi Hydropower Station. [Online]. Available: <http://www.nea.org.np>
- [2] Nepal Energy Forum. [Online]. Available: <http://www.nepalenergyforum.com>
- [3] HydroWorld. [Online]. Available: <http://www.hydroworld.com>

...